

# VARUN GOPAL

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## PROFESSIONAL SUMMARY

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Backend developer with 4 years of experience building high-performance REST APIs, optimizing MySQL at scale, and integrating third-party services in a healthcare technology environment. Strong track record in query optimization, stored procedures, and database performance tuning. Experienced working within microservices architectures, writing authentication middleware, and designing async data sync logic. Currently building on this backend foundation with coursework and self-directed projects in agentic AI systems and tool-augmented workflows using Google ADK. Looking to contribute backend engineering expertise while developing reliable, production-ready systems.

## TECHNICAL SKILLS

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**Languages & Frameworks:** Python, Django, Django REST Framework, Flask, JavaScript (React basics)

**AI & Agentic Tools:** Google ADK (Agent Development Kit), Gemini API, FastMCP (Model Context Protocol server), Multi-Agent Orchestration, RAG (Retrieval-Augmented Generation), Structured Outputs (Pydantic), Custom Function Tools, Google Search Grounding, Code Execution Tools

**Databases & Storage:** MySQL (stored procedures, functions, indexing, query optimization), ChromaDB (vector search basics), AWS S3

**Tools & Platforms:** Docker, Git, AWS (Lambda, CloudWatch), JWT Authentication, REST APIs, Multi-Tenant Architecture, Blue-Green Deployment, Linux, Cron Jobs, Jenkins CI/CD

**Development & Collaboration:** Postman, Swagger, Jira, Visual Studio Code, Pytest, Agile/Scrum

## PROFESSIONAL EXPERIENCE

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Senior Software Engineer | Equipo Health Inc. | Trivandrum, Kerala

June 2022 – July 2026

- **API & Backend Development:** Built and maintained REST APIs within an existing microservices architecture using Django REST Framework and MySQL. Developed endpoints for patient records, referrals, e-consultations, and clinical reporting — handling millions of rows with sub-100ms response targets.
- **Database Performance & Optimization:** Led query optimization and indexing efforts on high-traffic endpoints. Used EXPLAIN analysis to eliminate full table scans and filesorts; added composite and covering indexes. Moved complex reporting aggregations into MySQL stored procedures, reducing query parse overhead and network round-trips between Python and the database.
- **Authentication & Security:** Wrote JWT middleware and role-based permission checks that verified tokens locally on every request — no per-request database lookup needed. Scoped all patient and document queries by tenant (group/facility) extracted from the authenticated user's token, enforcing multi-tenant isolation at the application layer.
- **Healthcare Data Handling:** Built backend flows for clinical document uploads: validated file types and sizes, stored files in private S3 buckets, and persisted metadata in MySQL. Generated short-lived pre-signed S3 URLs for secure provider access, with authorization checks ensuring users only accessed their own patients' documents.
- **Real-Time Data Synchronization:** Designed async, queue-based sync logic to keep patient data consistent between the core application and an external care-management platform. Defined event triggers, duplicate-prevention rules (idempotent upserts by unique event ID), and conflict-resolution logic (provider-entered data takes precedence). Collaborated with DevOps on Lambda deployment and queue configuration.
- **Patient Engagement Automation:** Built scheduling and eligibility logic for automated patient outreach — appointment reminders, medication nudges, and follow-ups — delivered via SMS, IVR, and email through the Telnix API. Implemented timezone-aware delivery windows and database-level status locks to prevent duplicate sends.
- **Internal Control Panel:** Developed backend APIs for a React-based internal admin panel that let ops teams self-serve configuration changes — group/facility settings, referral routing rules, and user permissions. Added server-side pagination, filtering, and select\_related optimizations to prevent N+1 queries.
- **Monitoring & Logging:** Wrote custom Django middleware that tagged every request with a unique request ID, captured structured JSON logs (path, status code, duration, user ID), and attached the request ID to response headers, enabling direct tracing of user-reported issues in AWS CloudWatch Logs Insights.
- **Automation & Scheduled Jobs:** Built Python scripts for batch processing — fax generation, insurance document handling, reminder dispatch — scheduled through Jenkins, with database-level atomic status locks to prevent overlapping job runs.
- **DevOps Collaboration:** Developed features within Dockerized environments (multi-stage builds, non-root containers) and collaborated with DevOps on Jenkins-based CI/CD pipelines and blue-green deployments.

## PROJECTS

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### CivicPulse AI — Multi-Agent Civic Grievance & Escalation System (self-directed mini-project)

*Google ADK, FastMCP, Django REST, MySQL, ChromaDB, Gemini API*

- Designed a municipal grievance resolution system using Google ADK with a root coordinator agent that routes citizen queries to three specialist sub-agents (Triage, Knowledge, Escalation) based on intent, each with its own tool access and instructions.
- Built a FastMCP tool server over streamable HTTP exposing deterministic, raw-SQL-backed tools (list categories, geospatial duplicate detection via a custom haversine implementation, complaint filing, SLA-breach checks) so domain actions and SLA math are never left to LLM inference.
- Modeled a configurable SLA-rules table (per category and priority, with a default fallback) driving deadline computation, and implemented a bounding-box pre-filter plus precise haversine check for location-based duplicate detection.
- Built a RAG pipeline over the grievance charter using ChromaDB with a custom Gemini-embeddings function (asymmetric RETRIEVAL\_DOCUMENT/RETRIEVAL\_QUERY task types) for grounded, citation-backed policy answers.
- Implemented a before-model callback that regex-filters prompt-injection attempts and off-topic requests before they reach the LLM, and a before-tool callback that logs every tool call into session state as an audit trail.
- Bridged Django REST Framework's sync views to ADK's async Runner via a dedicated background asyncio event loop (thread-safe client, coroutine submission), with MySQL-backed session persistence (DatabaseSessionService) and separate fast-path ORM endpoints for status/stats polling that skip the LLM entirely.

### Google Skills Coursework — Agentic AI & Tool Integration

*Google ADK, Gemini API, Pydantic, Python*

- Practiced structured data extraction using Pydantic output schemas and session state templating for reliable extraction and state propagation across conversation turns.
- Built agents using custom function tools, Google Search grounding, and code execution tools.
- Configured agents using both programmatic Python setups and declarative YAML configurations matching ADK schema standards.

## EDUCATION

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### Bachelor of Engineering (B.E.) in Computer Science and Engineering

SNS College of Engineering, Anna University | Coimbatore, Tamil Nadu | 2014 – 2018

## CERTIFICATIONS

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- Engineer AI Agents with Agent Development Kit (ADK) – Google Cloud
- Create Your First Gemini Enterprise Application – Google Cloud
- SQL (Advanced) – HackerRank
- API Security Fundamentals – APIsec University
- Problem Solving (Intermediate) – HackerRank
- Data Processing and Visualization – Accenture (via SSC NASSCOM)

## LANGUAGES

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English (Professional Working Proficiency) • Malayalam (Native) • Tamil (Professional Working Proficiency)